

exercise-bank

Typst Package

Exercise Management & Banking System

A comprehensive solution for creating, organizing, and filtering exercises

Version 0.5.3

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1. Introduction

`exercise-bank` is a Typst package for creating and managing exercises with solutions, metadata, filtering, and exercise banks. Perfect for teachers, textbook authors, and educational content creators.

1.1. Features

- Exercises with inline or deferred solutions
- Teacher corrections with flexible display modes
- Draft mode for work-in-progress documents
- Multiple display control options (what, where, which content)
- Metadata support (topic, level, author, competencies)
- Exercise banks: define once, display anywhere
- Powerful filtering by any criteria
- Automatic numbering with reset options
- Customizable labels (localization support)

1.2. Installation

Import the package in your Typst document:

```
#import "@preview/exercise-bank:0.5.3": exo, exo-setup
```

1.3. Quick Start

Code

```
#exo(  
  exercise: [Solve the equation  $2x + 5 = 13$ .]  
)
```

Preview

Exercise 1 Solve the equation $2x + 5 = 13$.

2. Basic Usage

2.1. Simple Exercise

Code

```
#exo(  
  exercise: [Calculate $3 + 4 times 2$.]  
)
```

Preview

Exercise 1 Calculate $3 + 4 \times 2$.

2.2. Exercise with Solution

Code

```
#exo(  
  exercise: [Calculate $3 + 4 times 2$.],  
  solution: [$3 + 4 times 2 = 3 + 8 = 11$],  
)
```

Preview

Exercise 1 Calculate $3 + 4 \times 2$.

Solution 1 $3 + 4 \times 2 = 3 + 8 = 11$

2.3. Multiple Exercises

Exercises are automatically numbered:

Code

```
#exo(exercise: [Simplify $x^2 + 2x + 1$.])  
#exo(exercise: [Factor $x^2 - 4$.])  
#exo(exercise: [Solve $2x - 6 = 0$.])
```

Preview

Exercise 1 Simplify $x^2 + 2x + 1$.

Exercise 2 Factor $x^2 - 4$.

Exercise 3 Solve $2x - 6 = 0$.

3. Display Control

The package uses three parameters to control display behavior.

3.1. display - What to Display

Controls what content is displayed:

- "both" (default) - Show exercises and solutions/corrections
- "ex" - Show only exercises
- "sol" - Show only solutions/corrections

Code

```
#exo-setup(display: "ex") // Exercises only

#exo(
  exercise: [Solve  $x + 3 = 7$ .],
  solution: [ $x = 4$ ], // Hidden
)
```

Preview

Exercise 1 Solve $x + 3 = 7$.

3.2. corr-display - Which Content to Show

Controls whether to show solutions or corrections:

- "solution" (default) - Show solution content
- "correction" - Show correction content
- "mixed" - Default to solution, but use correction for exercises with `show-corr: true`

Code

```
#exo-setup(corr-display: "correction")

#exo(
  exercise: [Simplify  $2x + 3x$ .],
  correction: [
     $2x + 3x = 5x$ 
    (Combine like terms)
  ],
)
```

Preview

Exercise 1 Simplify $2x + 3x$.

Correction 1 $2x + 3x = 5x$ (Combine like terms)

3.3. corr-loc - Where to Display

Controls where solutions/corrections appear:

- "after" (default) - Immediately after each exercise
- "pagebreak" - With a page break between
- "end-section" - Collected at section end
- "end-chapter" - Collected at chapter end

Code

```
#exo-setup(corr-loc: "end-section")

#exo(exercise: [Exercise 1], solution: [Answer 1])
#exo(exercise: [Exercise 2], solution: [Answer 2])

#exo-print-solutions() // Print collected solutions
```

Preview

Exercise 1	Exercise 1
-------------------	------------

Exercise 2	Exercise 2
-------------------	------------

Solutions

Solution 1	Answer 1
-------------------	----------

Solution 2	Answer 2
-------------------	----------

4. Corrections (Teacher Version)

Corrections are detailed solutions for teachers, including pedagogical notes and teaching tips.

4.1. Exercise with Correction

Code

```
#exo-setup(corr-display: "correction")

#exo(
  exercise: [Solve  $x^2 = 9$ .],
  correction: [
    *Teacher notes:*
     $x = \pm 3$ 

    Common mistake: forgetting the negative root.
  ],
)
```

Preview

Exercise 1 Solve $x^2 = 9$.

4.2. Solutions Only (Answer Key)

Show only solutions for student answer keys:

Code

```
#exo-setup(display: "sol")

#exo(
  exercise: [This is hidden],
  solution: [Only this solution is shown],
)
```

Preview

Solution 1 Only this solution is shown

4.3. The sol-in-corr Flag

When `corr-display: "correction"`, both correction AND solution are shown by default. Use `sol-in-corr: true` on an exercise to indicate the solution is already embedded in the correction (avoiding duplication):

Code

```
#exo-setup(corr-display: "correction")

// Without sol-in-corr: both correction and solution shown
#exo(
  exercise: [Problem A],
  correction: [Teacher notes only],
  solution: [$x = 5$],
)

// With sol-in-corr: only correction shown
#exo(
  exercise: [Problem B],
  correction: [
    *Solution:* $x = 3$

    _Teaching tip: Watch for sign errors._
  ],
  solution: [$x = 3$],
  sol-in-corr: true, // Solution already in correction
)
```

Preview

Exercise 1 Problem A

Exercise 2 Problem B

4.4. Mixed Display Mode

Use `corr-display: "mixed"` to default to solutions while showing corrections for specific exercises:

Code

```
#exo-setup(corr-display: "mixed")

#exo(
  exercise: [Simple problem],
  solution: [Quick answer],
  correction: [Detailed explanation],
)

#exo(
  exercise: [Complex problem],
  solution: [Answer],
  correction: [Step-by-step solution],
  show-corr: true, // Shows correction
)
```

Preview

Exercise 1 Simple problem

Exercise 2 Complex problem

5. Draft Mode

Show placeholders for incomplete exercises during document preparation.

5.1. With Draft Mode ON

Code

```
#exo-setup(  
  draft-mode: true,  
  solution-placeholder: [_To be written_] ,  
)  
  
#exo(  
  exercise: [Solve  $x + 5 = 12$ ],  
  solution: [], // Empty - shows placeholder  
)
```

Preview

Exercise 1 Solve $x + 5 = 12$

5.2. With Draft Mode OFF (Default)

Empty solutions show minimal space without placeholders:

Code

```
#exo-setup(draft-mode: false)  
  
#exo(  
  exercise: [Solve  $x + 5 = 12$ ],  
  solution: [], // Empty - no placeholder  
)
```

Preview

Exercise 1 Solve $x + 5 = 12$

6. Exercise Banks

Define exercises once, display them anywhere. Perfect for creating reusable exercise collections.

6.1. Defining Bank Exercises

Use `exo-define` to register exercises without displaying them:

```
#exo-define(  
  id: "quad-1",  
  exercise: [Solve  $x^2 - 5x + 6 = 0$ .],  
  topic: "quadratics",  
  level: "1M",  
  solution: [ $x = 2$  or  $x = 3$ ],  
)  
  
#exo-define(  
  id: "geom-1",  
  exercise: [Find the area of a circle with radius 5.],  
  topic: "geometry",  
  level: "1M",  
  solution: [ $A = 25\pi$ ],  
)
```

6.2. Displaying Bank Exercises

Use `exo-show` to display a specific exercise:

Code

```
#exo-show("quad-1")
```

Preview

Exercise 1 Solve $x^2 - 5x + 6 = 0$.

6.3. Selecting Multiple Exercises

Use `exo-select` with filters:

```
// All quadratics exercises  
#exo-select(topic: "quadratics")  
  
// Level 1M exercises only  
#exo-select(level: "1M")  
  
// Multiple topics  
#exo-select(topics: ("quadratics", "geometry"))  
  
// Limit count  
#exo-select(topic: "algebra", max: 5)  
  
// Custom filter  
#exo-select(where: ex => ex.metadata.level == "hard")
```

7. Metadata and Filtering

7.1. Adding Metadata

Tag exercises for organization and filtering:

```
#exo(  
  exercise: [Solve  $x + 1 = 5$ .],  
  topic: "algebra",  
  level: "easy",  
  authors: ("Prof. Smith",),  
)
```

7.2. Filtering Exercises

Display only exercises matching criteria:

```
#exo-filter(topic: "algebra")  
#exo-filter(level: "easy")  
#exo-filter(topic: "algebra", level: "hard")
```

8. Competency Tags

Tag exercises with competencies and display them visually.

Code

```
#exo-setup(show-competencies: true)

#exo-define(
  id: "comp-ex",
  exercise: [Solve and explain your reasoning.],
  competencies: ("C1.1", "C2.3", "C4.1"),
  solution: [Detailed solution here],
)

#exo-show("comp-ex")
```

Preview

Exercise 1 Solve and explain your reasoning.

C1.1 C2.3 C4.1

8.1. Filter by Competency

```
#exo-select(competency: "C1.1")
#exo-select(competencies: ("C1.1", "C2.3"))
```

9. Configuration

9.1. Global Setup

Use `exo-setup` to configure defaults:

```
#exo-setup(  
  // Display control  
  display: "both",           // "ex", "sol", "both"  
  corr-display: "solution",  // "solution", "correction", "mixed"  
  corr-loc: "after",         // "after", "pagebreak", "end-section", "end-chapter"  
  // Labels  
  exercise-label: "Exercise",  
  solution-label: "Solution",  
  correction-label: "Correction",  
  // Counter behavior  
  counter-reset: "section",  // "section", "chapter", "global"  
  // Display options  
  show-id: false,  
  show-competencies: false,  
  // Draft mode  
  draft-mode: false,  
  // Badge styling  
  badge-style: "box",  
  badge-color: black,  
)
```

9.2. Badge Styles

Change the visual style of exercise badges:

```
// Circled number (no label text)  
#exo-setup(badge-style: "circled")  
  
// Filled circle with white number  
#exo-setup(badge-style: "filled-circle", badge-color: rgb("#2563eb"))  
  
// Pill-shaped badge  
#exo-setup(badge-style: "pill")  
  
// Arrow tag  
#exo-setup(badge-style: "tag", badge-color: rgb("#1e40af"))
```

Available styles: "box" (default), "circled", "filled-circle", "pill", "tag", "margin", "border-accent", "underline", "rounded-box", "header-card"

9.3. Localization

Change labels for different languages:

Code

```
// French  
#exo-setup(  
  exercise-label: "Exercice",  
  solution-label: "Solution",  
  correction-label: "Corrigé",  
)
```

Preview

French: Exercice, Solution, Corrigé
German: Aufgabe, Lösung
Spanish: Ejercicio, Solución

9.4. Counter Reset Options

Control when numbering resets:

```
#exo-setup(counter-reset: "section") // Reset each section
#exo-setup(counter-reset: "chapter") // Reset each chapter
#exo-setup(counter-reset: "global")  // Never reset
```

Use hooks to trigger resets:

```
= New Section
#exo-section-start() // Resets if counter-reset: "section"

= New Chapter
#exo-chapter-start() // Resets if counter-reset: "chapter"
```

9.5. Advanced Exercises

Mark exercises as advanced to display a visual cue before the label. The default symbol is *.

Code

```
#exo(
  exercise: [A challenging problem.],
  advanced: true,
)
```

Preview

*** Exercise 1** A challenging problem.

Customize the symbol:

Code

```
#exo-setup(advanced-symbol: sym.dagger)
#exo(
  exercise: [Advanced with dagger.],
  advanced: true,
)
```

Preview

† Exercise 1 Advanced with dagger.

Disable with `advanced-symbol: none`.

9.6. Optional Exercises

Mark exercises as optional to display the optional-star icon before the label. Students know they can skip these.

Code

```
#exo(
  exercise: [A bonus problem.],
  optional: true,
)
```

Preview

★ Exercise 1 A bonus problem.

Customize or disable the symbol via `exo-setup`:

```
#exo-setup(optional-symbol: [★]) // custom
#exo-setup(optional-symbol: none) // disable
```

9.7. Correction-Given Exercises

The dumbbell icon (`corr-given: true`) signals that the printed correction will be handed out to students, so they can work independently with the answer key.

Code

```
#exo(  
  exercise: [Factor  $x^2 - 9$ .],  
  solution: [ $(x-3)(x+3)$ .],  
  corr-given: true,  
)
```

Preview

↗ Exercise 1 Factor $x^2 - 9$.

Customize via `exo-setup(corr-given-symbol: ...)` or disable with `none`.

10. QR Codes

Attach a QR code to any exercise (e.g. linking to a video correction or an online resource). Pass a URL string — the code is generated with `tiaoma` — or ready-made content via the `qr` parameter.

Code

```
#exo(  
  exercise: [Factorise  $x^2 - 9$ .],  
  solution: [ $(x-3)(x+3)$ .],  
  qr: "https://example.com/eq1",  
)
```

Preview

Exercise 1 Factorise $x^2 - 9$.

The QR code is placed automatically based on the active badge style:

- **Badge styles** (`box`, `circled`, `filled-circle`, `pill`, `tag`, `margin`): the QR sits below the badge in the label margin.
- **Full-width styles** (`border-accent`, `underline`, `rounded-box`, `header-card`): the exercise content wraps around the QR at the top right.

Solution and correction boxes can carry their own separate QR codes via `qr-sol` and `qr-corr`.

10.1. Global QR Settings

Control QR appearance globally via `exo-setup`:

Code

```
#exo-setup(  
  qr-size: 1.5cm,  
  qr-color: rgb("#1e3a8a"),  
  qr-caption: [Correction],  
  show-qr: true,  
)
```

Preview

Use `show-qr: false` to suppress all QR codes (e.g. for a print version) without removing the URLs from bank definitions.

11. Visual Styles

Set the badge style with `exo-setup(badge-style: "...")`.

11.1. Box (Default)

Code

```
#exo-setup(badge-style: "box")
#exo(exercise: [Solve  $x + 3 = 7$ ])
```

Preview

Exercise 1 Solve $x + 3 = 7$

11.2. Circled

Code

```
#exo-setup(badge-style: "circled")
#exo(exercise: [Solve  $x + 3 = 7$ ])
```

Preview

1 Solve $x + 3 = 7$

11.3. Filled Circle

Code

```
#exo-setup(
  badge-style: "filled-circle",
  badge-color: rgb("#2563eb"),
)
#exo(exercise: [Solve  $x + 3 = 7$ ])
```

Preview

1 Solve $x + 3 = 7$

11.4. Pill

Code

```
#exo-setup(badge-style: "pill")
#exo(exercise: [Solve  $x + 3 = 7$ ])
```

Preview

Exercise 1 Solve $x + 3 = 7$

11.5. Tag

Code

```
#exo-setup(
  badge-style: "tag",
  badge-color: rgb("#1e40af"),
)
#exo(exercise: [Solve  $x + 3 = 7$ ])
```

Preview

Exercise 1 Solve $x + 3 = 7$

11.6. Border Accent

Code

```
#exo-setup(
  badge-style: "border-accent",
  badge-color: rgb("#3b82f6"),
)
#exo(exercise: [Solve  $x + 3 = 7$ ])
```

Preview

Exercise 1
Solve $x + 3 = 7$

11.7. Underline

Code

```
#exo-setup(badge-style: "underline")
#exo(exercise: [Solve  $x + 3 = 7$ ])
```

Preview

Exercise 1

Solve $x + 3 = 7$

11.8. Rounded Box

Code

```
#exo-setup(badge-style: "rounded-box")
#exo(exercise: [Solve  $x + 3 = 7$ ])
```

Preview

Exercise 1

Solve $x + 3 = 7$

11.9. Header Card

Code

```
#exo-setup(badge-style: "header-card", badge-color: rgb("#3b82f6"))
#exo(exercise: [Solve  $x + 3 = 7$ ])
```

Preview

Exercise 1

Solve $x + 3 = 7$

11.10. Margin

The label appears flush-right in a fixed-width left column; content flows in the right column. Solutions are shown in a compact bordered block.

Code

```
#exo-setup(badge-style: "margin")
#exo(
  exercise: [Solve  $x + 3 = 7$ ],
  solution: [ $x = 4$ ],
)
```

Preview

Exercise 1: Solve $x + 3 = 7$

12. Parameter Reference

12.1. `exo` Function

Parameter	Type	Default	Description
<code>exercise</code>	content	none	Exercise content (required)
<code>solution</code>	content	none	Solution content
<code>correction</code>	content	none	Correction for teachers
<code>id</code>	string	auto	Unique exercise ID
<code>margin-content</code>	content	none	Content placed below the badge (e.g. remarks)
<code>qr</code>	string/content	none	QR code for the exercise box (URL string or content)
<code>qr-sol</code>	string/content	none	QR code for the solution box
<code>qr-corr</code>	string/content	none	QR code for the correction box
<code>sol-in-corr</code>	bool	false	If true, solution is already in correction (don't show both)
<code>show-corr</code>	bool	false	If true, show correction in “mixed” mode
<code>optional</code>	bool	false	Show the optional marker before the label
<code>corr-given</code>	bool	false	Show the correction-given marker (dumbbell icon)
<code>topic</code>	string	none	Topic metadata
<code>level</code>	string	none	Difficulty level
<code>authors</code>	array	()	Author names

12.2. `exo-define` Function

Same as `exo`, plus:

Parameter	Type	Default	Description
<code>optional</code>	bool	false	Show the optional marker before the label
<code>corr-given</code>	bool	false	Show the correction-given marker (dumbbell icon)
<code>competencies</code>	array	()	Competency tags
<code>points</code>	number	none	Points (for exam mode)

12.3. `exo-select` Function

Parameter	Type	Default	Description
<code>topic</code>	string	none	Filter by topic
<code>level</code>	string	none	Filter by level
<code>author</code>	string	none	Filter by author
<code>competency</code>	string	none	Filter by competency
<code>topics</code>	array	none	Filter by any topic

levels	array	none	Filter by any level
competencies	array	none	Filter by any competency
where	function	none	Custom filter function
renumber	bool	true	Renumber sequentially
max	int	none	Maximum exercises

12.4. `exo-setup` Function

Display Control:

Parameter	Type	Default	Description
display	string	“both”	“ex”, “sol”, “both”
corr-display	string	“solution”	“solution”, “correction”, “mixed”
corr-loc	string	“after”	“after”, “pagebreak”, “end-section”, “end-chapter”
exercise-label	string	“Exercise”	Label for exercises
solution-label	string	“Solution”	Label for solutions
correction-label	string	“Correction”	Label for corrections
counter-reset	string	“section”	“section”, “chapter”, “global”
show-id	bool	false	Show exercise IDs
show-competencies	bool	false	Show competency tags
draft-mode	bool	false	Show placeholders for empty content
solution-placeholder	content	<i>To be completed</i>	Placeholder text
correction-placeholder	content	<i>To be completed</i>	Placeholder text

Visual Styling:

Parameter	Type	Default	Description
badge-style	string	“box”	Style: “box”, “circled”, “filled-circle”, “pill”, “tag”, “margin”, “border-accent”, “underline”, “rounded-box”, “header-card”
badge-color	color	black	Color for exercise badges
solution-color	color	green	Color for solution badges
correction-color	color	green	Color for correction badges
label-font-size	length	12pt	Font size for badge labels
margin-position	length/auto	auto	Width reserved for badge column
label-extra	length	1cm	Extra space for labels in margin

Spacing:

Parameter	Type	Default	Description
exercise-above	length	0.8em	Space above exercise boxes

exercise-below	length	0.8em	Space below exercise boxes
solution-above	length	0.8em	Space above solution boxes
solution-below	length	0.8em	Space below solution boxes
correction-above	length	0.8em	Space above correction boxes
correction-below	length	0.8em	Space below correction boxes
advanced-symbol	content/none	"*"	Symbol before label for advanced exercises
optional-symbol	content/none	star icon	Symbol before label for optional exercises
corr-given-symbol	content/none	dumbbell icon	Symbol before label when correction is handed out

QR Code Settings:

Parameter	Type	Default	Description
show-qr	bool	true	Master toggle for per-exercise QR codes
qr-size	length	1.5cm	Target QR code size (shrinks to fit margin if needed)
qr-min-size	length	1cm	Minimum size; below this the QR extends into page margin
qr-color	color	black	QR module color
qr-caption	content/none	none	Small caption below every QR code
qr-position	string	"auto"	"auto" (per badge style) or "wrap" (always wrap content)

13. Utility Functions

Function	Description
<code>exo-reset-counter()</code>	Reset exercise numbering to 0
<code>exo-clear-registry()</code>	Clear all registered exercises
<code>exo-section-start()</code>	Call at section start (triggers reset if configured)
<code>exo-chapter-start()</code>	Call at chapter start (triggers reset if configured)
<code>exo-print-solutions()</code>	Print collected solutions (for end-section/chapter modes)
<code>exo-count(topic: ..)</code>	Count exercises matching criteria
<code>exo-show("id")</code>	Display exercise by ID
<code>exo-show-many("a", "b")</code>	Display multiple exercises by ID
<code>exo-filter(topic: ..)</code>	Filter and display matching exercises

14. Exam Integration

These functions support building exam documents from an exercise bank, typically used alongside a layout package such as `texam`.

14.1. exam-setup

Configure the exam correction box appearance and behaviour.

Parameter	Type	Default	Description
<code>show-solutions</code>	bool	false	Show <code>exam-solution-box</code> content
<code>solution-label</code>	string	"Correction"	Label shown in bold at the top of the solution box
<code>solution-fill</code>	color	green tint	Background fill of the solution box
<code>solution-stroke</code>	stroke	green stroke	Border stroke of the solution box
<code>solution-radius</code>	length	3pt	Corner radius of the solution box
<code>solution-inset</code>	length	8pt	Inner padding of the solution box
<code>solution-label-color</code>	color	dark green	Color of the label text
<code>question-spacing</code>	length	1em	Vertical space between questions

14.2. exam-solution-box

A green-highlighted answer block. Only rendered when `exam-setup(show-solutions: true)`.

```
#exam-solution-box[
  $(x - 2)(x - 3)$
]
```

Place it after an exercise. The same source file produces both the student version (box hidden) and the correction (box shown) by flipping `show-solutions`.

14.3. exam-score-table

Builds a score grid from registered exercise IDs, reading point values from the registry.

```
#exam-score-table("ex-1", "ex-2", "ex-3")
```

Parameter	Type	Default	Description
<code>..exercise-ids</code>	strings		Positional — IDs of exercises to include
<code>question-label</code>	string	"Question"	Header for the question column
<code>points-label</code>	string	"Points"	Header for the points row
<code>grade-label</code>	string	"Note"	Header for the grade row
<code>total-label</code>	string	"Total"	Label for the total column